Janghyun **Noh**

AI Researcher · Data Scientist

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Education

Myongji University — *Seoul, S.Korea*

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Thesis: *Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data*
- Advisor: Prof. Minsik Oh, Artificial Intelligence and Data Analytics (AIDA) Lab
- GPA: 4.44 / 4.5
- Award: Research Honor Award from the President of Myongji University (at graduation)
- Teaching Assistant: Deep Learning, Machine Learning, Data Mining, Basic Programming (Fall 2024 – Spring 2026)

Myongji University — *Seoul, S.Korea*

B.S. in Data Technology, School of Convergence Software — *Mar. 2018 - Aug. 2024*

- Capstone Project: CatchYou, a generative AI service that creates composite sketches from witness descriptions
- Leadership: Secretary General, Student Council

RED School – Alternative School — *Geumsan, S.Korea*

Student — *Mar. 2012 - Jan. 2018*

- Scholarship: Academic Excellence Scholarship (two consecutive years)
- Leadership: Student Council President

Experience

Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Graduate Researcher — *Sep. 2024 - Aug. 2026*

- Developed scCap, a deep learning framework that predicts patient phenotypes from single-cell data without manual cell type annotation, using knowledge-augmented clustering (first-author paper in Briefings in Bioinformatics).
- Developed a multimodal fusion framework for COVID-19 severity prediction that combines clinical records, chest X-rays, and radiology reports and is extensible to new modalities (with KDCA).
- Developed deep learning models that classify tumor subtypes from single-cell data and applied them to drug target discovery (with AIgenDrug).
- Built an LLM-based AI agent environment (Biomni) to automate genomic data analysis for a CRISPR genome-editing analysis platform.
- Set up a scalable genome analysis workflow for apple whole-genome data using the Nextflow nf-core/sarek pipeline (with KNU).
- Analyzed spatial gene-expression data from breast cancer tissue to identify cell neighborhoods and the tumor microenvironment (with SNUH).
- Compared gene-expression patterns between liver cancer tissues with and without cirrhosis using spatial transcriptomics data.

Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Undergraduate Researcher — *Dec. 2022 - Aug. 2024*

- Developed machine learning models to predict in-hospital mortality of ARDS patients from ICU records, comparing missing-data strategies.
- Identified hypertension-progression risk groups in hyperlipidemia patients from KoGES cohort data by designing lab-based labels and building machine learning and deep learning models (with KDCA).

Bigleader — *Seoul, S.Korea*

AI Intern (Youth AI Internship) — *Jul. 2022 - Oct. 2022*

- Built a supply-chain risk monitoring system that automatically collects foreign customs notices, and validated its keyword extraction and HS code matching (with KOTRA).
- Received the KOTRA Merit Award (President's Award) and the TriBIG Project Scholarship.

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data — *Briefings in Bioinformatics (BIB)*

Published — *Jul. 2026*

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids — *Journal of Experimental & Clinical Cancer Research (JECCR)*

Published — *May 2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq
KIISE Korea Computer Congress (KCC)
Oral Presentation — *Jun. 2026*
Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques
KIISE Korea Software Congress (KSC)
Poster Presentation (Excellent Poster Award) — *Dec. 2025*
Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data
International Conference on Genome Informatics ISCB-Asia
Poster Accepted — *Dec. 2025*
J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning
BIOINFO 2025, Korean Society for Bioinformatics (KSBI)
Poster Presentation (Excellent Poster Award) — *Oct. 2025*
J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning
KIISE Korea Computer Congress (KCC)
Oral Presentation (Excellent Presentation Award) — *Jul. 2025*
J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies
KIISE Korea Software Congress (KSC)
Poster Presentation — *Dec. 2023*
J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2026	**Research Honor Award (President's Award),** Myongji University	*Seoul, S.Korea*
2025	**Excellent Poster Award,** KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award,** BIOINFO 2025, Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Excellent Presentation Award,** KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**Merit Award (President's Award),** Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship,** Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship,** Myongji University	*Seoul, S.Korea*
2018	**Scholarship III,** Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship,** RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship,** RED School	*Geumsan, S.Korea*

Projects

2024 – 2026	AI-based target discovery platform for CAR-Treg cell therapy in chronic inflammatory disease
2024 – 2026	Genetic and physiological mechanisms of apple blossom dormancy release under climate change
2024 – 2026	Spatial transcriptomics-based target discovery and validation for refractory solid tumors
2025 – 2026	Client-side web platform for detecting large CRISPR-Cas9-induced DNA deletions
2025 – 2026	GeoMx spatial transcriptomics of cirrhotic vs. non-cirrhotic HCC tissue
2025	COVID-19 severity prediction using multimodal clinical and imaging data
2023 – 2024	Hypertension-progression risk groups in hyperlipidemia patients (KoGES cohort)
2023	CatchYou: criminal composite sketch generation from witness descriptions
2022	Customs notice analysis and monitoring system for global supply-chain crises

Technical Skills

Programming	Python, R
Frameworks	PyTorch, TensorFlow, Scikit-learn
Tools	Linux, Git, Docker, Nextflow
Research	Machine Learning, Deep Learning, Multimodal Learning
Languages	Korean (Native), English (Fluent)
Certifications	Engineer Information Processing (2023), SQL Developer – SQLD (2023)

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University
Seoul, S.Korea

Member — *Oct. 2022 - Dec. 2023*

- Joined the AI team and engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Data Youth Campus, MSIT & KDATA
Seoul, S.Korea

Participant — *Jul. 2022 - Aug. 2022*

- Completed a government-funded data science training program run by the Ministry of Science and ICT and KDATA, hosted at Dongguk University
- Conducted practical machine learning and data analysis projects

DeepsDeeps, Myongji University
Seoul, S.Korea

Member — *Mar. 2022 - Jul. 2022*

- Studied deep learning fundamentals and paper implementations in a peer-led deep learning study group

FAA Football Analytics Academy, Bepro
Seoul, S.Korea

Participant — *Sep. 2021 - Dec. 2021*

- Completed the Data Integration Course on integrating and analyzing football match data

Coding 101, Myongji University
Seoul, S.Korea

Member — *Mar. 2018 - Jan. 2020*

- Studied programming fundamentals and worked on coding projects in a student club

International

Camino de Santiago Pilgrimage
Spain

Pilgrim — *Sep. 2026 - Oct. 2026*

- Walked the French Way on foot from Saint-Jean-Pied-de-Port to Santiago de Compostela

Medical Mission Trip
Manila, Philippines

Volunteer — *Jan. 2024*

- Supported medical volunteer activities with international healthcare professionals through the Global Medical Missions Alliance (GMMA)

International Work Camp
Reykjavik, Iceland

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and intercultural exchange with international volunteers

Short-term Mission Trip
Hanoi, Vietnam

Volunteer — *Aug. 2018*

- Participated in intercultural exchange and community volunteer activities